

1. Consider the equation $x = 2$.
 - (a) In $\mathbb{R}$, represent geometrically the set of points that satisfy the equation.
 - (b) In $\mathbb{R}^2$, represent geometrically the set of points that satisfy the equation.
 - (c) In $\mathbb{R}^3$, represent geometrically the set of points that satisfy the equation.
2. Consider the equation $x^2 = x$.
 - (a) In $\mathbb{R}$, represent geometrically the set of points that satisfy the equation.
 - (b) In $\mathbb{R}^2$, represent geometrically the set of points that satisfy the equation.
 - (c) In $\mathbb{R}^3$, represent geometrically the set of points that satisfy the equation.
3. Consider the equation $2x + y = 3$.
 - (a) In $\mathbb{R}^2$, represent geometrically the set of points that satisfy the equation.
 - (b) In $\mathbb{R}^3$, represent geometrically the set of points that satisfy the equation.
4. Consider the equation $y = x^2 + 1$.
 - (a) In $\mathbb{R}^2$, represent geometrically the set of points that satisfy the equation.
 - (b) In $\mathbb{R}^3$, represent geometrically the set of points that satisfy the equation.
5. Consider the equation $y = x + 1$.
 - (a) In $\mathbb{R}^2$, represent geometrically the set of points that satisfy the equation.
 - (b) In $\mathbb{R}^3$, represent geometrically the set of points that satisfy the equation.
6. Consider the points $P = (1, 0, 2)$, $Q = (1, 1, 3)$ e $R = (-1, 1, 0)$ em $\mathbb{R}^3$.
 - (a) Determine $\overrightarrow{PQ} \times \overrightarrow{PR}$
 - (b) Write the equation of the plane passing through points P, Q and R .
7. Consider the equation $x^2 + y^2 = 2$.
 - (a) In $\mathbb{R}^2$, represent geometrically the set of points that satisfy the equation.
 - (b) In $\mathbb{R}^3$, represent geometrically the set of points that satisfy the equation.
8. Sketch in $\mathbb{R}^3$ the quadric surfaces:
 - (a) $x^2 + 2y^2 + \frac{z^2}{9} = 1$
 - (b) $x^2 + z^2 = 9$
 - (c) $x^2 + y^2 = z^2$
 - (d) $z = x^2 + y^2$
 - (e) $z = x^2 + y^2 - 6$
 - (f) $z = x^2 - y^2$
 - (g) $x^2 - y^2 - z^2 = 9$
 - (h) $x^2 + y^2 - z^2 = 9$
 - (i) $z = \sqrt{4 - x^2 - y^2}$
 - (j) $z = \sqrt{x^2 + y^2}$.

9. In $\mathbb{R}^3$, represent geometrically the set of points that satisfy the equation $x^2 + 4y^2 = 4$.
10. Consider the surface with equation $z^2 = x^2 + y^2$.
- (a) Determine the intersection of the given surface with each of the coordinate planes.
 - (b) Determine the intersection of the given surface with the planes $x = 1$, $y = 1$ and $z = 1$.
 - (c) Sketch the graph of the given surface.